II. Say True or False.

1. Enzyme substrate complex is a permanent stable complex.
2. Malonate is the competitive inhibitor of succinate dehydrogenase.
3. An enzyme substrate complex is formed in all the enzymatic reactions.
4. The degree of competitive inhibition cannot be decreased by increasing the concentration of the substrate.
5. An uncompetitive inhibitor has affinity towards ES complex.

III. Match the following

1. Enzymes - $\frac{1}{V}$ vs $\frac{1}{S}$
2. ES complex – affinity of enzyme for substrate
3. k_m - Unstable and highly energetic
4. Line weaver Burk plot - Biocatalysts

IV. Give short answers for the following

1. Define K_m
2. What is the nature of active site according to lock and key theory ?
3. What is competitive inhibition ?
4. What is induced fit theory ?
5. What is irreversible enzyme inhibition ?

V. Answer the following

1. Derive MM equation.
2. How is the LineWeaver Burk plot arrived at?
3. Explain the concept of competitive inhibition
4. What is the action of malonate on succinate dehydrogenase ?
5. Compare competitive and non-competitive inhibition.

XII U9 Enzyme Kinetics.indd 226 4/8/2020 3:27:16 PM